Informatica

Informatica® Cloud Data Integration

Microsoft Dynamics NAV Connector

Informatica Cloud Data Integration Microsoft Dynamics NAV Connector
December 2017

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Preface

Use *Data Integration Microsoft Dynamics NAV Connector* to learn how to read from or write to Microsoft Dynamics NAV by using Cloud Data Integration. Learn to create a Microsoft Dynamics NAV connection and run synchronization tasks in Cloud Data Integration.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

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- View product availability information.
- Create and review your support cases.
- Find your local Informatica User Group Network and collaborate with your peers.

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Informatica Marketplace

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To find online support resources on the Informatica Network, click **Contact Support** in the Informatica Intelligent Cloud Services Help menu to go to the **Cloud Support** page. The **Cloud Support** page includes system status information and community discussions. Log in to Informatica Network and click **Need Help** to find additional resources and to contact Informatica Global Customer Support through email.

The telephone numbers for Informatica Global Customer Support are available from the Informatica web site at https://www.informatica.com/services-and-training/support-services/contact-us.html.

CHAPTER 1

Introduction to Microsoft Dynamics NAV Connector

This chapter includes the following topics:

- Microsoft Dynamics NAV Overview, 6
- NAV Server Security Policy Configuration, 7
- How to Publish NAV Pages, 10
- Different Types of Page Services, 11
- User Access and Permission Setup, 12
- Assumptions and Limitations, 12

Microsoft Dynamics NAV Overview

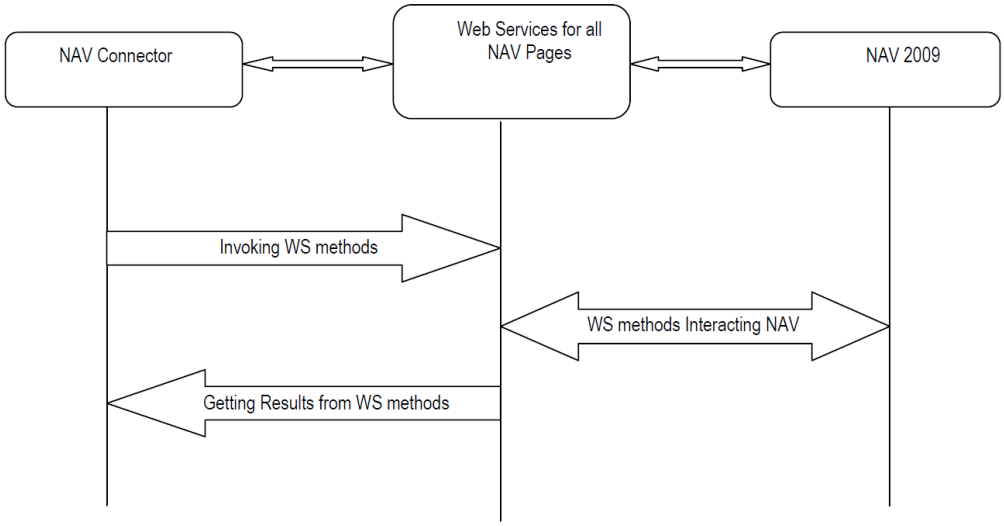
Microsoft Dynamics NAV Connector enables you to securely read data from or write data to Microsoft Dynamics NAV. Microsoft Dynamics NAV sources and targets represent records in Microsoft Dynamics NAV.

Microsoft Dynamics NAV Connector uses JDK 1.6, and MS Dynamics NAV 2009 SP1/R2 technologies.

Microsoft Dynamics NAV Connector uses JAX-WS to run the web service methods dynamically.

Every object in NAV 2009 is exposed as web service with standard methods for reading data from or writing data to NAV 2009.

The following image displays the architectural diagram of Microsoft Dynamics NAV Connector:



NAV Server Security Policy Configuration

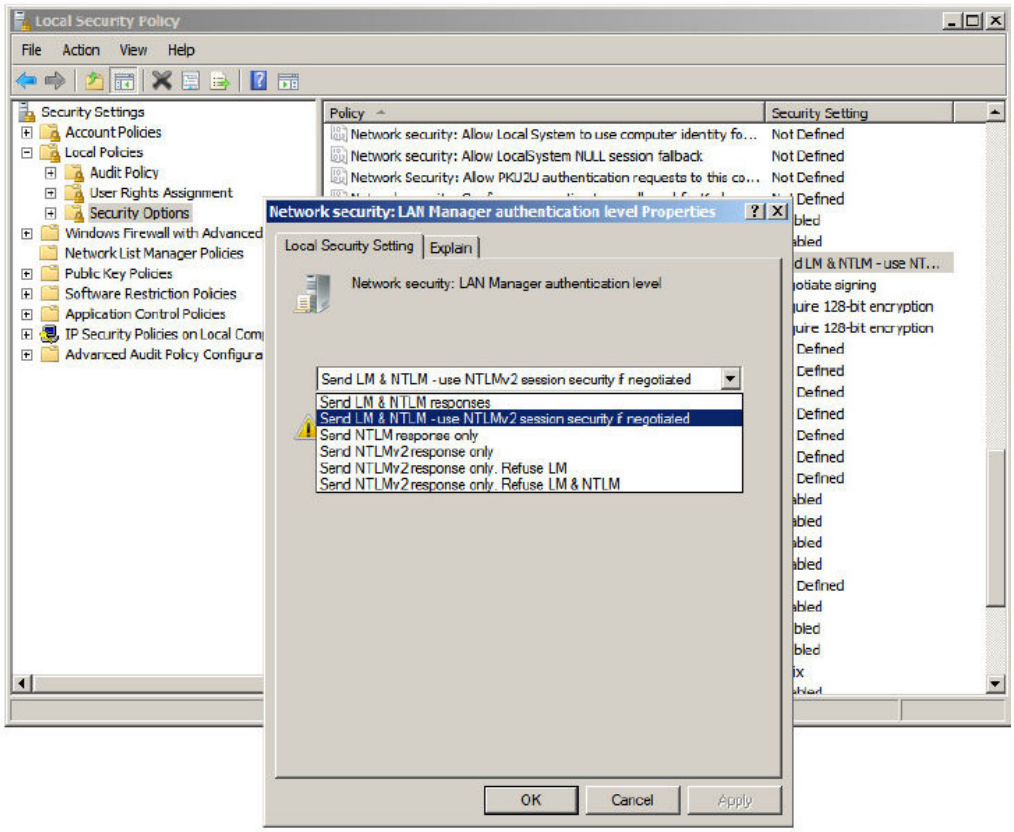
Perform the following steps to configure the NAV Server Security Policy:

1. Open **Local Security Policy** page.

   Run **secpol.msc**.

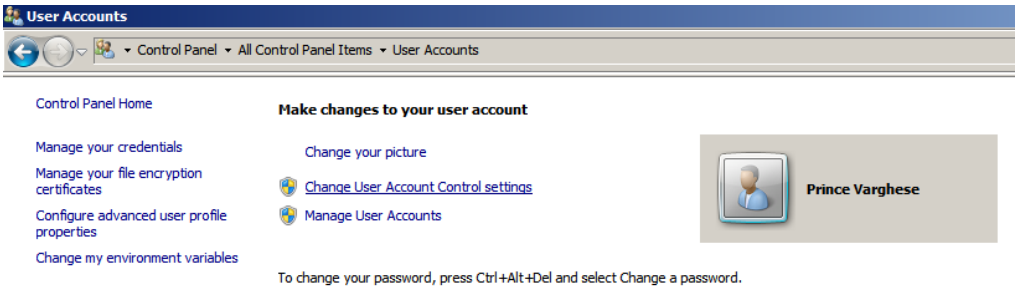
2. Go to **Local Policies** > **Security Options**.
3. Open **Network security: Lan Manager authentication level Properties** and select **Send LM & NTLM -use NTLMv2 session security if negotiated**.

The following image displays the local security policies window:



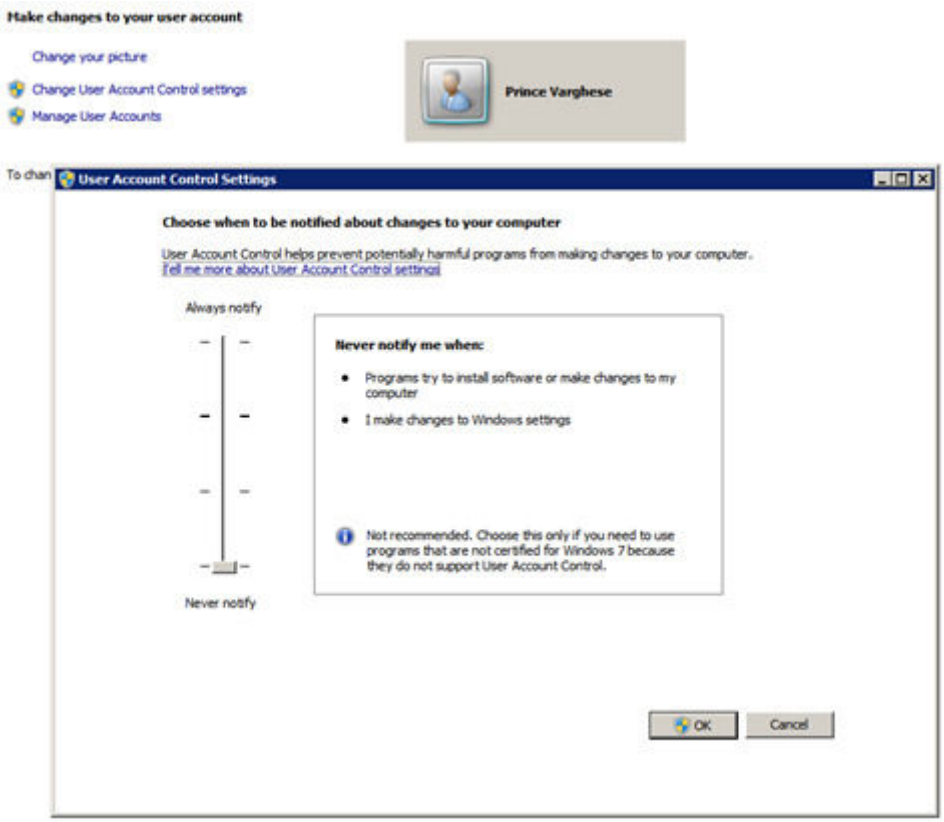
4. Click **Apply** and then **OK**.
5. Go to **Control Panel** > **All Control Panel Items** > **User Accounts** > **Change User Account Control settings**.

   The following image displays the user accounts window:



6. Change the slider to Never Notify.

   The following image displays user account control settings window:



7. Restart the machine and NAV service.

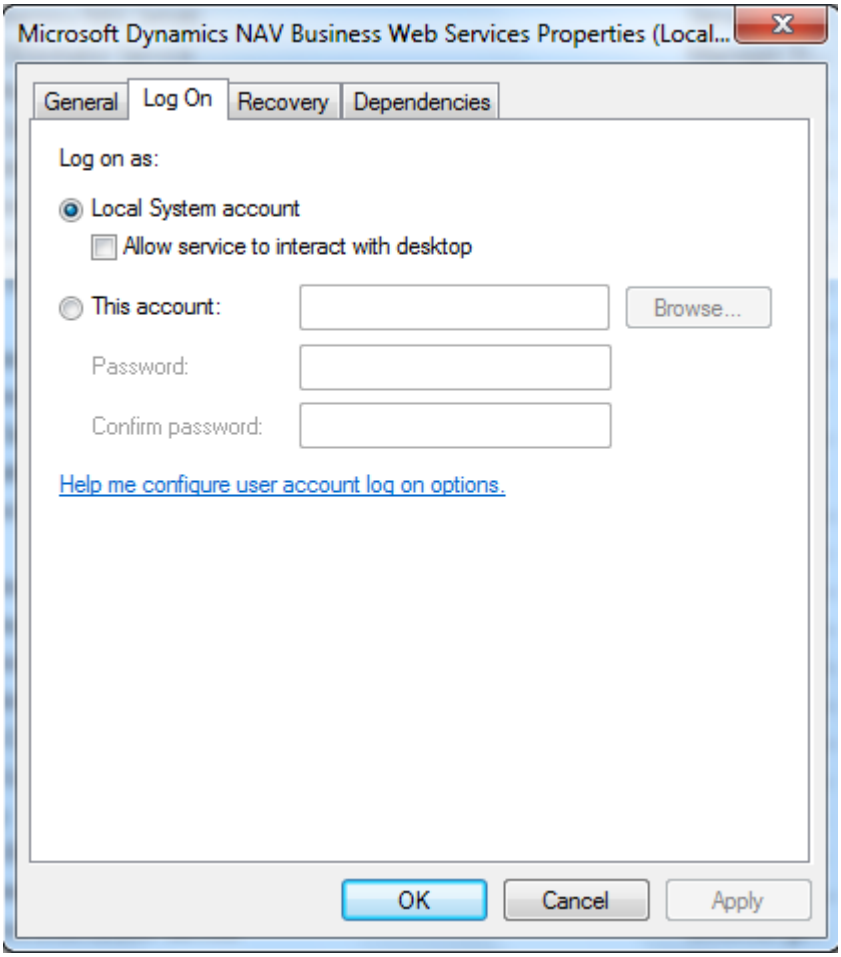
NAV Web Service Configuration

You can use Microsoft NAV web service to integrate with Microsoft Dynamics NAV.

Perform the following steps to configure NAV web services:

1. Select **Microsoft Dynamics NAV Business Web Services Properties** Service in NAV server page.

2. Select **Local System account** on **Log On** tab in **Microsoft dynamics NAV Business Web Services Properties** window.



How to Publish NAV Pages

Perform the following steps to publish NAV pages:

1. Run **810 Web Services** form in the **Object Designer** page.
2. Go to **Page** in the **Object Type** tab.

   Select the appropriate page ID and provide a name to the **Page Service** page.

   Note: Append List for List type Pages and Card for Card type Pages.
3. Enable the **Published** option.
4. Restart the **Microsoft Dynamics NAV Business Web Services** service.
5. Start Internet Explorer.
6. In the **Address** field, enter the following address where the keys are replaced with the values specified in the `CustomSettings.config` file: `http://<Server>:<WebServicePort>/<ServerInstance>/WS/<CompanyName>/services`. For example,

```
http://localhost:7047/DynamicsNAV/WS/CRONUS%20International%20Ltd./services
```
The page must list the Web service that you just published.

7. Close the browser.

Different Types of Page Services

Microsoft Dynamics NAV Connector supports the following types of page services:

- Card Type Page Services
  Card Type Page services support both insertion and reading process with the pages. Card Type Page services have methods to read, write, update, and delete. The following list displays the methods that Card Type Page services support:

Operation	Methods supported
Read	Read()
	ReadMultiple()
Write	Create()
	CreateMultiple()
Update	Update()
	UpdateMultiple()
Delete	Delete()

- List Type Page Services
  List Type Page Services support read operation. List type page services retrieve the records based on the index values that have been maintained by SQL Server. It support Read() and ReadMultiple().

  Note: Card type page services and list type page services do not support all the fields in the SQL table. Card type pages have fields that accept values from the user. List type pages have auto-calculated fields or system fields that reads or displays data.

  The following table lists the object type, name, source table, and total fields:

Object Type	Name	Source Table	Total Fields
Table	Customer		55 (Approx)
Page (Card)	CustCard	Customer	35 (Input Type and Auto Fill Type Fields)
Page (List)	CustList	Customer	20 (Display purpose Fields)

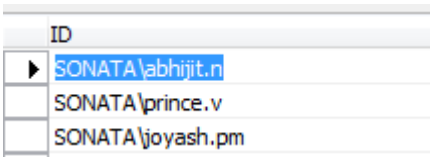
  You can add a field when you design the page and publish the page. The new field is available in the WSDL schema.

User Access and Permission Setup

Perform the following steps to access Microsoft Dynamics NAV.

1. Go to **Tools** > **Security** > **Windows Login** in NAV web page.
2. Create an Entry under the ID field with a valid user ID.
3. Select **Roles**.
4. Specify Super to grant the SUPER role permission to the user.

The following image displays a sample user ID:



Assumptions and Limitations

The following list describes the assumptions and limitations of Microsoft Dynamics NAV Connector:

- The user must be a domain user and must be the Windows Login list to access the Navision Web Service.
- The user can perform read and write operations by using card type pages. The list type pages supports only read operations.
- When you read data from NAV, the size of the data that can be handled depends on the string variable limit.
- When you update the records, you must remove the system fields such as Last_date_modified from field mapping.

Microsoft Dynamics NAV Connector supports the following basic filters:

- INSERT
  The field name key must be present at the parent level. If a child key is present, the field name KEY_1 must be present at the child level. You do not need values for the key field to run an insert operation. If the key field has a value, the insert operation ignores key field values and inserts a new record in the database.

  A parent entity can have multiple child entities. A parent record can have multiple child records.

  When you insert a child record into a parent record, the parent record must have an appropriate value for the key field. The child record must have null value for the key field. If the fields are auto populated when you insert entities, you must remove the auto populated fields.

  For example, if you provide the value for the field Sell_to_Customer_No in SalesInvoicePage, then the value for the fields Sell_to_Customer_Name, Sell_to_Customer_Contact_No, Sell_to_Address, Sell_to_Post_Code generates automatically using Sell_to_Customer_No value. Hence, remove the fields Sell_to_Customer_Name, Sell_to_Customer_Contact_No, Sell_to_Address, Sell_to_Post_Code when you insert data into SalesInvoicePage.
- UPDATE
  When you invoke an update operation, the values for key field must be present for both parent and child record.
- UPSERT

If the record does not have a value for the key field, the record is inserted in to the database. If the record has a value for the key field, the record is updated. If the value of the key field does not match, an error message appears.

- DELETE
  You must provide both parent key and child key to delete a record.

CHAPTER 2

Microsoft Dynamics NAV Connections

This chapter includes the following topics:

- Microsoft Dynamics NAV Connections Overview, 14
- Microsoft Dynamics NAV connection properties, 14
- Page Service Methods and Parameters , 15

Microsoft Dynamics NAV Connections Overview

You can use Microsoft Dynamics NAV connection to read data from or write data to Microsoft Dynamics NAV. You can use Microsoft Dynamics NAV connections to specify sources and targets in synchronization tasks.

Create a connection and associate it with a synchronization task, mapping, or mapping task. Define the source and target properties to read data from or write data to Microsoft Dynamics NAV.

You create a Microsoft Dynamics NAV connection on the **Connections** page. You can then use the connection in the Mapping Designer when you create a mapping or in the Synchronization Task wizard when you create a synchronization task.

Microsoft Dynamics NAV connection properties

When you set up a Microsoft Dynamics NAV connection, you must configure the connection properties.

Important: Microsoft Dynamics NAV Connector is deprecated and has been moved to maintenance mode. Informatica intends to drop support in a future release.

The following table describes the Microsoft Dynamics NAV connection properties:

Property	Description
Runtime Environment	The name of the runtime environment where you want to run the tasks.
Username	User name of the NAV account.
Password	Password for the NAV account.
Host Name	Name of the NAV Server host.
Port	NAV web service port number.
Service Instance	Name of the Microsoft Dynamics NAV Server instance for web service.
Company Name	Name of the company in NAV to which the user belongs.
Domain_Name	Domain name.

Page Service Methods and Parameters

The following table lists the description, prototype, and return type for page service methods and parameters:

Operation	Description	Prototype	Return Type
Read	To read a single record based on the key field value as a parameter. For example, CustPage CustPage_Service.Read(string No)	PageType PageService.Read(string Value)	PageType
Readmultiple	To read a set of records based on the specified record length in the parameter list	PageType[] PageService.ReadMultiple(PageType_Flter[] filter, string BookmarkKey, int setSize)	PageType[]
Create	Create a new record into the published page (Actual insertion happening inside the SQL table)	Void PageServie.Create(ref PageTypeInstance PageType)	Void
Createmultiple	To create multiple records	Void PageService.CreateMultiple(ref PageTypeInstance[] PageTypeList)	Void
Update	To update a single record with new values (Similar to CREATE method)	Void PageService.Update(ref PageTypeInstance PageType)	Void

Operation	Description	Prototype	Return Type
Updatemultiple	To update multiple records in a single execution (Similar to CREATEMULTIPLE method)	Void PageService.UpdateMultiple(ref PageTypeInstance[] PageTypeList)	Void
Delete	To delete a single record from the page (Actual deletion happening inside the SQL table)	Boolean PageService.Delete(string Key) (Key Value can be accessible from the KEY Field from the correspondence table)	Void

CHAPTER 3

Synchronization Tasks with Microsoft Dynamics NAV Connector

This chapter includes the following topic:

- Connector Read and Write Notes, 17

Connector Read and Write Notes

Read

Microsoft Dynamics NAV Connector reads the data from the company that is specified in the connection. Records are processed based on the company. You can apply filters to the data. READMULTI is the NAV web service method that reads the data.

Write

1. NAV web service method used to insert is – CREATE
2. NAV web service method used to delete is – DELETE
3. NAV web service method used to insert is – UPDATE
4. NAV web service methods used to upsert is – UPDATE & INSERT

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